

Johnson C. Smith University

Calculus II | MTH 232 | Section A | Spring 2013

“Every act of conscious learning requires the willingness to suffer an injury to one’s self-esteem. That is why young children, before they are aware of their own self-importance, learn so easily; and why older persons, especially if vain or important, cannot learn at all.”

-Thomas Szasz, author, professor of psychiatry (1920-)

Instructor Information

Name: Dr. James Ashe

Office location: 307 Davis Hall (back); office hours will also be held in the Math Lab, EDU 115

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Office hours: MWF 10–11am and 12-1pm; By appointment only, MW 8-9am and 2-3pm.

Course Description

This is a **3 credit hour** course; the second of a sequence of four calculus courses which are offered for students from the sciences, social sciences and engineering. Topics covered: Anti-derivatives, Fundamental Theorem of Calculus and definite integrals, Applications of the definite integral to area, volume, force and work, and arc length. Transcendental functions. Techniques of integration. Prerequisite: MTH 231 (Calc I).

Instructional methods will primarily consist of lectures and in-class discussions, however other methods may be adopted according to the particular needs of the class.

Course Objectives

Students should be able to demonstrate competency in the techniques and topics outlined in the section above. Additionally, students should understand the meaning of the integral (both as a limit of Riemann sums and as the net accumulation of the rate of change), the relationship between the derivative and integral, and develop problem solving skills.

Course Materials

Textbook: *Calculus, Briggs and Cochran. ISBN: 978-0321336118*

Calculator: A TI-83, TI-83+, or TI-84+ are allowed and supported. Any other calculator must first be approved by the instructor. Students should bring their calculator to class.

Grades

Grading scale: $A \geq 90 > B \geq 80 > C \geq 70 > D \geq 60 > F$

Grades will be taken from homework, quizzes, and Exams as follows:

Evaluation: HW/Quizzes = 20%; Exams 1, 2, 3, and 4 = 15% each; Final Exam = 20%.

HW/Quizzes: HW/Quizzes will consist of any graded assignment which is not an exam. This will include traditional quizzes, take home assignments, and in-class projects. The lowest quiz grade will be dropped.

Exams: Dates for exams will announced at least one week in advance.

Final Exam: The final will be a cumulative exam. The final will be given in our normal class room on Thursday, May 16 from 3-5pm; see the exam schedule:

www.jcsu.edu/happenings/exam-schedule/spring-exam. *If beneficial, the final exam score will replace the lowest exam score.*

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student’s responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence. In such cases, *it is the student's responsibility to make arrangements to make up any work that will be missed before the assignment/quiz/exam is missed.*

Missed HW/quizzes: There will be no make-ups for in-class quizzes unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a make-up will be given depending on circumstances. Late assignments will have 25% deducted for each day it is late.

Missed exams: Missed exams may be made up at the discretion of the Instructor. (1) you must notify me by email or phone within 24 hours of the missed exam AND (2) you must have no more than 3 unexcused absences in the class. If BOTH of these conditions are met, a make-up will be given. However unless a documented excuse is provided, 10% will automatically be deducted from the make-up. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

jcsu.edu/admissions/future_students/meet_your_counselor, phone (704) 378-1044.

Course Content

Chapter 4: Applications of the Derivative

4.8 Antiderivatives

Chapter 5: Integration

5.1 Approximating areas under curves

5.2 Definite Integrals

5.3 Fundamental Theorem of Calculus

5.4 Working with Integrals

5.5 Substitution Rule

Chapter 6: Applications of Integration

6.1 Velocity and Net Change

6.2 Regions Between Curves

6.3 Volume by Slicing

6.5 Length of Curves

Chapters 7: Logarithmic and Exponential Functions

7.1 Inverse Functions

7.2 The Natural Logarithmic and Exponential Functions

7.3 Logarithmic and Exponential Functions with other Bases

Chapter 8: Integration Techniques

8.1 Integration by parts

8.2 Trigonometric Integrals

8.3 Trigonometric Substitution

8.4 Partial Fractions

8.7 Improper Integrals

8.8 Introduction to Differential Equations

Important Dates

February 18, 2013	Exam 1
March 11, 2013	Exam 2
April 12, 2013	Exam 3
May 6, 2013	Exam 4
May 15, 2013	Final Exam

With exception of the final, the above exam dates are subject to change. JCSU's Fall 2013 academic calendar can be found here: jcsu.edu/happenings/academic-calendar, and the exam schedule can be found here: jcsu.edu/happenings/exam-schedule

While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.